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Ders 1

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$\int_G \vec{V} \cdot d\vec{r}$ this physically use full. Work =

$dW = \vec{F} \cdot d\vec{r}$

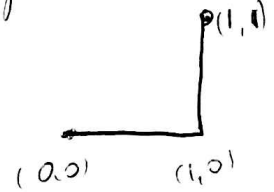
$= \int_G dW$. finds the total work done by $\vec{F}(x,y,z)$ along the contour, G

ex

Consider a function $\vec{F}(x,y,z) = 2xy\,\hat{x} + (x^2+y^2)\hat{y}$

find work done by $\vec{F}$. ($dW = \vec{F} \cdot d\vec{r}$).

along a contour



$\int_C \vec{F}(x,y,z) \cdot dr = \int \vec{F}_x(x,y,z)\hat{x}\,dx + (\dots) + (\dots)$.

$d\vec{r} = \hat{x}\,dx + \hat{y}\,dy + \hat{z}\,dz$

$W = \int_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F}_0 \cdot dr_0 + \int_{C_2} \vec{F} \cdot dr$

$\int_1 [2xy\,\hat{x} + (x^2+y^2)\hat{y}] \cdot (\hat{x}\,dx) = \int_0^1 2xy\,dx = yx^1\Big|_0^1 = (y) = 0$ *

$\int_0^1 (x^2+y^2)\,dy = \int_0 y^2 + 1\,dy = \frac{y^3}{3} + y\Big|_0^1 = \left(\frac{4}{3}\right)$

gaga em ?

Ders 2

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$$\oint_S \frac{\hat{r}}{r^2} \cdot d\vec{\sigma} = 0.$$

$S = S_{out} + S_{in} + S_{tunnel}$ Where S_{out} is surface of interest.

$$\int_{S_{out}} \frac{\hat{r}}{r^2} \cdot d\vec{\sigma} + \int_{S_{in}} \frac{\hat{r}}{r^2} \cdot d\vec{\sigma} + \underbrace{\int_{S_{tunnel}} \frac{\hat{r}}{r^2} d\vec{\sigma}}_{0} = 0$$

$$\int_{S_{in}} \frac{\hat{r}}{r^2} d\vec{\sigma} \qquad \text{for } S_{in} = d\vec{\sigma} = dA \cdot (-\hat{r})$$

r and δ relation!

$$-\int_{S_{in}} \frac{\hat{r}}{r^2} \cdot dA\,\hat{r} = -\int \frac{dA}{\delta^2} = -\frac{1}{\delta^2} \int dA = -\frac{1}{\delta^2} 4\pi\delta^2 = \boxed{-4\pi}$$

then

$$\int_S \frac{\hat{r}}{r^2} \cdot d\vec{\sigma} + \underbrace{\int_{S_{in}} \frac{\hat{r}}{r^2} \cdot d\vec{\sigma}}_{-4\pi} + \int_{S_{tunnel}} \frac{\hat{r}}{r^2} \cdot d\vec{\sigma} = 0$$

$$\boxed{\oint_{S\,arbitrary} \frac{\hat{r}}{r^2} \cdot d\vec{\sigma} = 4\pi \quad \text{if } S \text{ arbitrary encloses the origin.}}$$

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Thus

$$\vec{E} = \frac{q}{4\pi\epsilon_0}\vec{A} \Rightarrow \oint_S \vec{E}\cdot d\vec{\sigma} = \frac{q}{4\pi\epsilon_0}\cdot 4\pi = \boxed{\frac{q}{\epsilon_0}}$$

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2. Deo

$$\oint_S \vec{E}\cdot d\vec{\sigma} = \int_V (\nabla\cdot\vec{E})\, d\tau$$

$\frac{q}{\epsilon_0} = \int_V (\nabla\cdot\vec{E})\, d\tau$ = where q is sum off all charges in volume V enclosed by surface S.

$q = \int_V \rho\cdot d\tau$ where ρ is charge density.

$\underbrace{\rho\, d\tau}_{dq}$

$$\int_V \frac{\rho}{\epsilon_0} d\tau = \int_V \nabla\cdot\vec{E}\, d\tau \quad \text{for any Volume } V. \Rightarrow \boxed{\nabla\cdot E = \frac{\rho}{\epsilon_0}}$$

$\vec{E} = -\nabla\varphi_E$ where φ_E is electric potential.

divergence of gradient.

$$\nabla\cdot(-\nabla\varphi_E) = \frac{\rho}{\epsilon_0} \Rightarrow \nabla\cdot\nabla\varphi = -\frac{\rho}{\epsilon_0} \Rightarrow \boxed{\nabla^2 V = \frac{-\rho}{\epsilon_0}}$$

poisson, equation